



PUNJAB PUBLIC SERVICE COMMISSION
COMBINED COMPETITIVE EXAMINATION
FOR RECRUITMENT TO THE POSTS OF
PROVINCIAL MANAGEMENT SERVICE, ETC -2022
CASE NO. 2C2023

SUBJECT: MATHEMATICS (PAPER-I)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
- ii. Write Q. No. in the Answer Book in accordance with Q. No. in the Q. Paper.
- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
- iv. Extra attempt of any question or any part of the question will not be considered.

NOTE:

Attempt FIVE Questions in All including THREE questions from Part-A and TWO questions from Part-B. Calculator is allowed. (Non-Programmable)

PART-A

Q.No.1 (a) For what values of a , m , and b does the function

$$f(x) = \begin{cases} 3, & x = 0 \\ -x^2 + 3x + a, & 0 < x < 1 \\ mx + b, & 1 \leq x \leq 2 \end{cases}$$

satisfy the hypotheses of the Mean Value Theorem on the interval $[0, 2]$?

(b) Find the asymptotes of the curve $2xy + 2y = (x - 2)^2$.
(10+10=20 Marks)

Q.No.2

a) Evaluate $\int \frac{\sin x}{\sin 3x} dx$

b) The velocity of a car travelling on the Motorway at 15 minutes intervals is as follows:

Time in hours, $t =$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$	$\frac{5}{4}$
Velocity in km/hr, $v(t) =$	100	90	115	120	80

Find the total distance travelled by the car in the 60-minute period from $t = \frac{1}{4}$ to $t = \frac{5}{4}$.

Q.No.3

(a) Find the area of the region bounded by the curve $y = 4x - x^2$, the x -axis, and the lines $x = 1$ and $x = 3$.

(b) Using rectangular rule for $n = 4$, approximate the value of the definite integral

$$\int_0^1 \frac{dx}{1+x^2}$$

(10+10=20 Marks)

Q.No.4

a) Discuss the motion of a particle moving in a straight line if it starts from rest at a distance a from a point O and moves with an acceleration equal to μ times its distance from O .

b) A culture initially has P_0 number of bacteria. At $t = 1$ hour the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , determine the time necessary for the number of bacteria to triple.

(10+10=20 Marks)

Q.No.5

(a) Solve the differential equation $\frac{dy}{dx} = (-2x + y)^2 - 7$; $y(0) = 0$.

(b) A particle of mass m oscillates in a line with natural period $\frac{2\pi}{\omega}$. If an applied force $F \cos pt$ now acts in the line so that the particle is instantaneously at rest at zero time at a distance d from the centre of oscillation, prove that the displacement of the particle from the centre at subsequent time t is $d \cos \omega t + \frac{F(\cos pt - \cos \omega t)}{(\omega^2 - p^2)m}$.

(10+10=20 Marks)

PART-B

Q.No.6

(a) The n th term of a sequence is $n^{1/n}$. Determine whether sequence converges or diverges.

(b) Determine the convergence or divergence of the series $1 + \frac{1 \cdot 2}{1 \cdot 3} + \frac{1 \cdot 2 \cdot 3}{1 \cdot 3 \cdot 5} + \frac{1 \cdot 2 \cdot 3 \cdot 4}{1 \cdot 3 \cdot 5 \cdot 7} + \dots$ by applying any appropriate test.

(10+10=20 Marks)

Q.No.7

a) Prove that the Necessary and sufficient condition for a function $W = f(Z) = U(x, y) + iV(x, y)$ to be an analytic function is that the four partial derivatives U_x, U_y, V_x, V_y exist, are continuous and satisfy the Cauchy Riemann equations at each point of D_f i.e., $\frac{\partial U}{\partial x} = \frac{\partial V}{\partial y}$ and $\frac{\partial U}{\partial y} = -\frac{\partial V}{\partial x}$.

b) Prove that the function $f(Z) = \sqrt{|xy|}$ is not analytic at the origin although the Cauchy Riemann equations are satisfied at the origin.

(10+10=20 Marks)

Q.No.8

(a) Find the equation of the osculating plane to the Helix

$$x = a \cos \theta, \quad y = b \sin \theta, \quad z = b\theta \quad \text{at } \theta = \frac{\pi}{2}$$

(b) Show that the radius of curvature ρ and radius of torsion σ of the curve

$$\vec{r} = (a \cos u, a \sin u, a \cos 2u) \text{ at } u = \frac{\pi}{4} \text{ are } \rho = 5a \text{ and } \sigma = \frac{5a}{6}.$$

(10+10=20 Marks)



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SUBJECT: MATHEMATICS (PAPER-II)

TIME ALLOWED: THREE HOURS

MAXIMUM MARKS: 100

NOTE:

- i. All the parts (if any) of each Question must be attempted at one place instead of at different places.
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- iii. No Page/Space be left blank between the answers. All the blank pages of Answer Book must be crossed.
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NOTE: Attempt any FIVE questions. Calculator is allowed. (Non-Programmable)

- Q.No.1** (a) If order of an element b is n . Then show that the elements $b^0, b^1, b^2, \dots, b^{n-1}$ are all distinct and $b^k = e$ iff k is divisible by n .
 (b) Show that the only idempotent element in a group G is its identity.
(10+10=20 Marks)

- Q.No.2** (a) Prove that every sub group of a cyclic group is itself cyclic.
 (b) Show that any two cyclic groups of the same order are isomorphic to each other.
(10+10=20 Marks)

- Q.No.3** (a) Define an integral domain. If p is a prime number, then show that ring of integers mod p is an integral domain.
 (b) Let F be a field of real numbers. Then set of all real valued functions whose n th derivative exist for $n = 1, 2, \dots$, is a subspace of all real valued continuous function on $[0, 1]$.
(10+10=20 Marks)

- Q.No.4** (a) Let U and W be 2-dimensional subspaces of $\mathbb{R}^3$. Show that $U \cap W \neq \{0\}$.
 (b) Give at the least three examples of infinite dimensional vector spaces.
(10+10=20 Marks)

- Q.No.5** (a) Show that the intersection of any number of topologies is also a topology and is coarser than each of the given topologies.
 (b) Show that a subspace of a topological space is itself a topological space.
(10+10=20 Marks)

- Q.No.6** (a) Solve

$$\begin{aligned} \frac{1}{4}x_1 + \frac{2}{4}x_2 + \frac{1}{4}x_3 &= 40 \\ \frac{2}{4}x_1 + \frac{1}{4}x_2 + \frac{1}{4}x_3 &= 50 \\ \frac{2}{4}x_1 + \frac{2}{4}x_3 &= 60 \end{aligned}$$

- (b) Show that the system $Ax = b$ has a unique solution if A is non-singular matrix.

(12+8=20 Marks)

Q.No.7

(a) Prove that

$$\begin{vmatrix} 1+a & 1 & 1 & 1 \\ 1 & 1+b & 1 & 1 \\ 1 & 1 & 1+c & 1 \\ 1 & 1 & 1 & 1+d \end{vmatrix}$$

$$=abcd \left(1 + \frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d}\right)$$

(b) Find the eigenvalues and corresponding eigenvectors of the matrix

$$\begin{bmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{bmatrix}$$

(10+10=20 Marks)

Q.No.8

(a) Find the solution of the given system of equations by reducing it to reduced echelon form

$$6x_1 - 6x_2 + 6x_3 = 6$$

$$2x_1 - 4x_2 - 6x_3 = 12$$

$$10x_1 - 5x_2 + 5x_3 = 30$$

(b) For what Value of λ do the following homogeneous equations have non trivial solutions? Find these solutions

$$(3 - \lambda)x_1 - x_2 + x_3 = 0$$

$$x_1 - (1 - \lambda)x_2 + x_3 = 0$$

$$x_1 - x_2 + (1 - \lambda)x_3 = 0$$

(10+10=20 Marks)



CSS VISTA

WITH SADIA ZAHOOR
& ALI HASSAN SARGANA